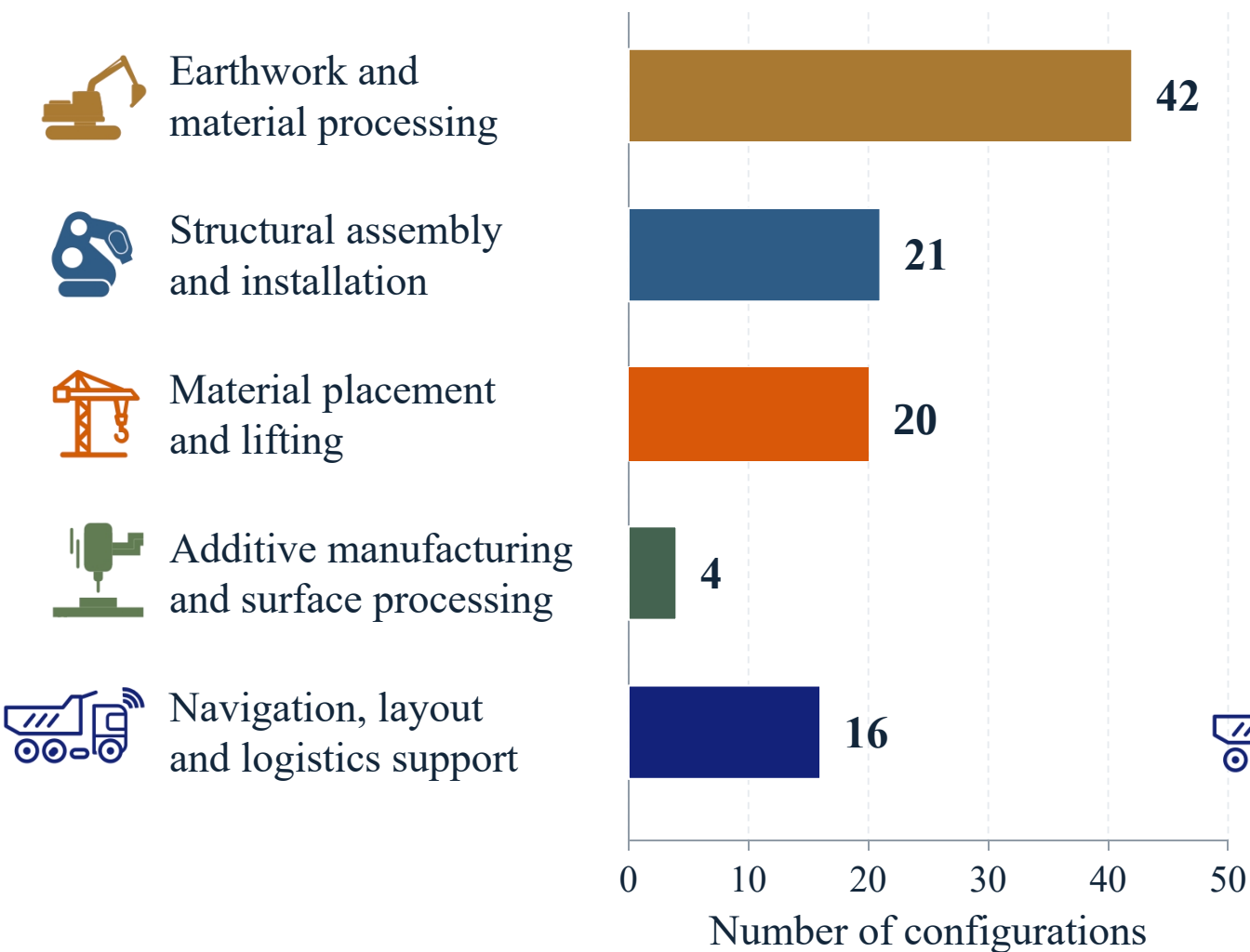


A Configurations by construction regime

Total number of configurations (N = 103)

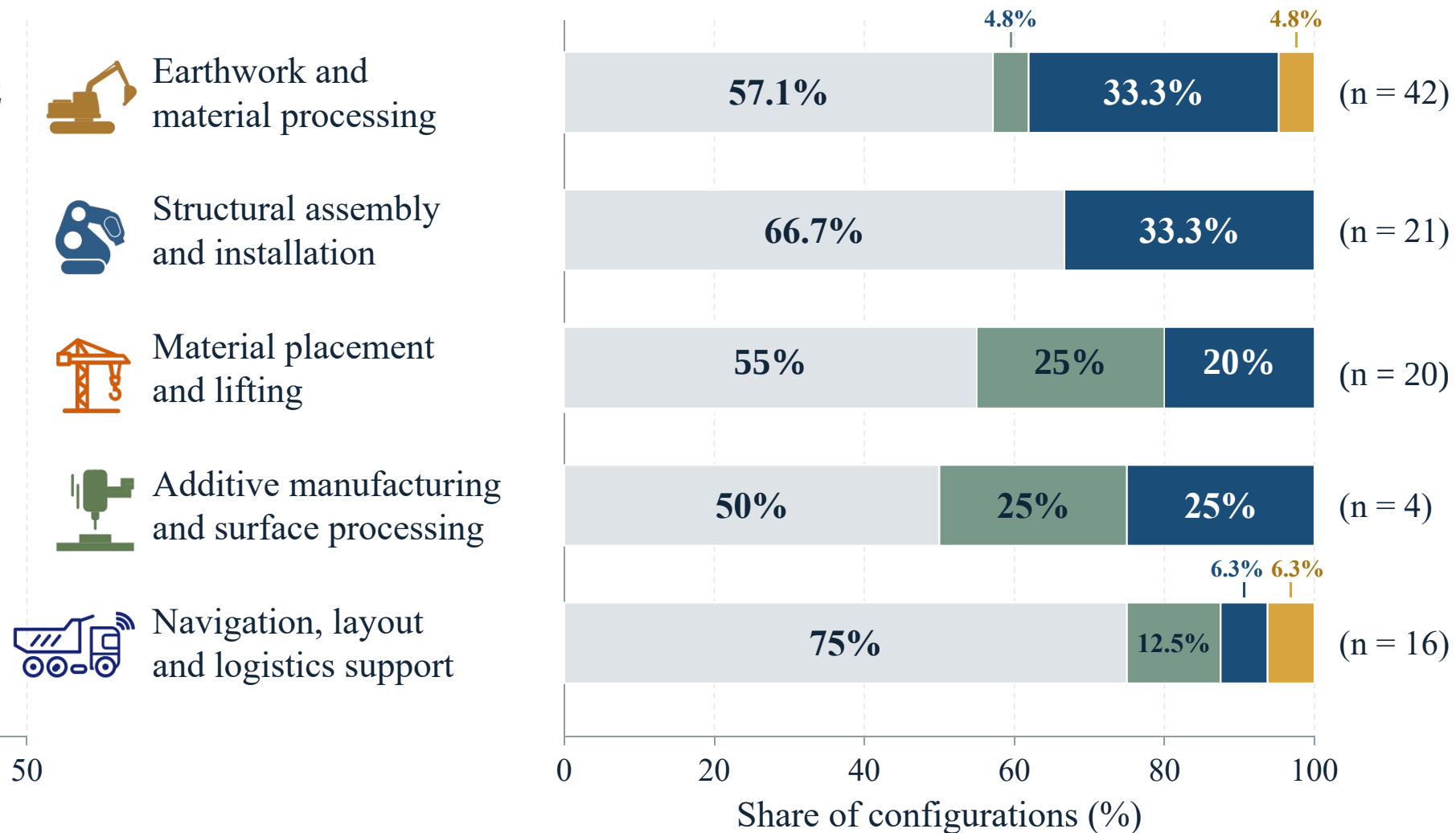


B Validation exposure within each regime

Share of configurations (%)

30 / 103 reached L3-L5

3 / 103 reached L5



Simulation (L0)

Virtual environments;
no closed-loop hardware

Laboratory hardware (L1-L2)

Isolated or varied
laboratory conditions

Construction testbeds / field trials (L3-L4)

Construction-representative testbeds
or near-operational field trials

Repeated workflows (L5)

Workflow-integrated
repeated operation

N = 103